

JHS IB/GES Curriculum

Sustainability & Environment • Grade 8 Term 2 Unit Demonstration

PORTFOLIO NOTE

This demonstration expands upon the example Grade 8, Term 2 unit documented in UET's JHS IB/GES curriculum artifact. It shows how the curriculum architecture can become a coherent, buyer-facing learning experience.

UNIT QUESTION

How can communities in Ghana reduce environmental harm while using science, data, communication and technology to develop practical solutions?

LEARNING OBJECTIVES

Investigate a local environmental or water-pollution issue using evidence.

Collect, organize and interpret relevant data.

Evaluate practical approaches to environmental protection, purification or conservation.

Communicate evidence-based recommendations to an authentic audience.

Collaborate across subjects to design a practical prototype or campaign.

INTERDISCIPLINARY DESIGN MAP

Science — investigate water pollution and/or test purification methods → investigation notes and findings.

Mathematics — collect and analyze rainfall or related environmental data → table/chart and interpretation.

Social Studies — examine environmental policies and community practices → policy/community brief.

Language — build a persuasive case for reducing plastic waste or protecting water → persuasive text.

Creative Arts — create a visual campaign, poster or short performance → campaign artifact.

Technology — develop a conceptual rainwater-harvesting solution → prototype and explanation.

INQUIRY SEQUENCE

Engage → Investigate → Analyze → Create → Share → Reflect.

LEARNER-FACING CHALLENGE

Choose a community environmental problem. Investigate what is happening, gather evidence, and develop a practical response. Your final presentation should explain what is happening, what evidence supports your understanding, what can be done, and why your proposed response is practical.

ASSESSMENT EVIDENCE

Research/investigation evidence; accurate or clearly interpreted data; feasible solution, prototype or campaign; clear communication to an audience; individual reflection on learning and ATL skills.

INSTRUCTOR GUIDANCE

Keep the inquiry evidence-based rather than steering learners toward a predetermined answer. Teams may investigate different environmental problems. Where physical resources are limited, teachers can use simulations, prepared datasets, photographs, offline research or low-cost materials.

REFLECTION / EXIT TICKET

What did the evidence help you understand?

What changed in your thinking or solution after investigation?

Which ATL skill did you use most?

What would you investigate or improve next?

UET DESIGN SIGNATURE

Inquiry → Evidence → Interdisciplinary application → Creation → Communication → Reflection.

EVIDENCE BOUNDARY

This is a UET portfolio demonstration based on the documented example unit in the original curriculum artifact. It does not claim school implementation, measured outcomes, accreditation, or formal authorization to deliver an IB programme.

Source: Original UET artifact, "MSA JHS IB Curriculum .pdf." Public portfolio title: JHS IB/GES Curriculum.